Example 1.35 (Discussion)

Now, since H is the incenter of DEF, we know

$$\angle APQ = 180^\circ - \angle AFQ = \angle BFQ = \angle BFD = \angle AFE = \angle AFP = \angle AQP$$

as desired. ■

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728 posts

Nov 16, 2021, 2:09 AM #9

ik im late but Solution (for practice)

> $\measuredangle AP_2Q_2 = \measuredangle AP_2B = \measuredangle ACB = \measuredangle ECD = \measuredangle EHD = \measuredangle EHA = \measuredangle EFA = \measuredangle P_2FA = \measuredangle P_2Q_2A$, thus we conclude that $AP_2 = AQ_2$.
> This is probably very similar to the written solution (I'd read it 2 months back) but I wanted to write in on my own, thus the post.

How come this solution did not require the conclusion that spoiler though? Have I inadvertently concluded that in my solution?

This post has been edited 1 time. Last edited by Hypatia1728, Nov 16, 2021, 2:11 AM

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May 13, 2022, 8:16 PM #10

> *ike.chen wrote:*
> Finally decided to write-up my original (not good) solution to the first ever "real" problem I solved from EGMO 🙂.
>
> Let H the the orthocenter of ABC.
>
> **Claim:** $AFQP$ is cyclic.
>
> **Proof.** Notice
>
> $$\angle ABH = \angle FBH = \angle FDH = \angle FDA = \angle ADF$$
>
> so $AFD \sim AHB$. Thus,
>
> $$\angle AFQ + \angle APQ = \angle AFD + \angle APB$$
>
> $$= \angle AHB + \angle ACB = 180^\circ$$
>
> by well-known properties of the orthocenter. □